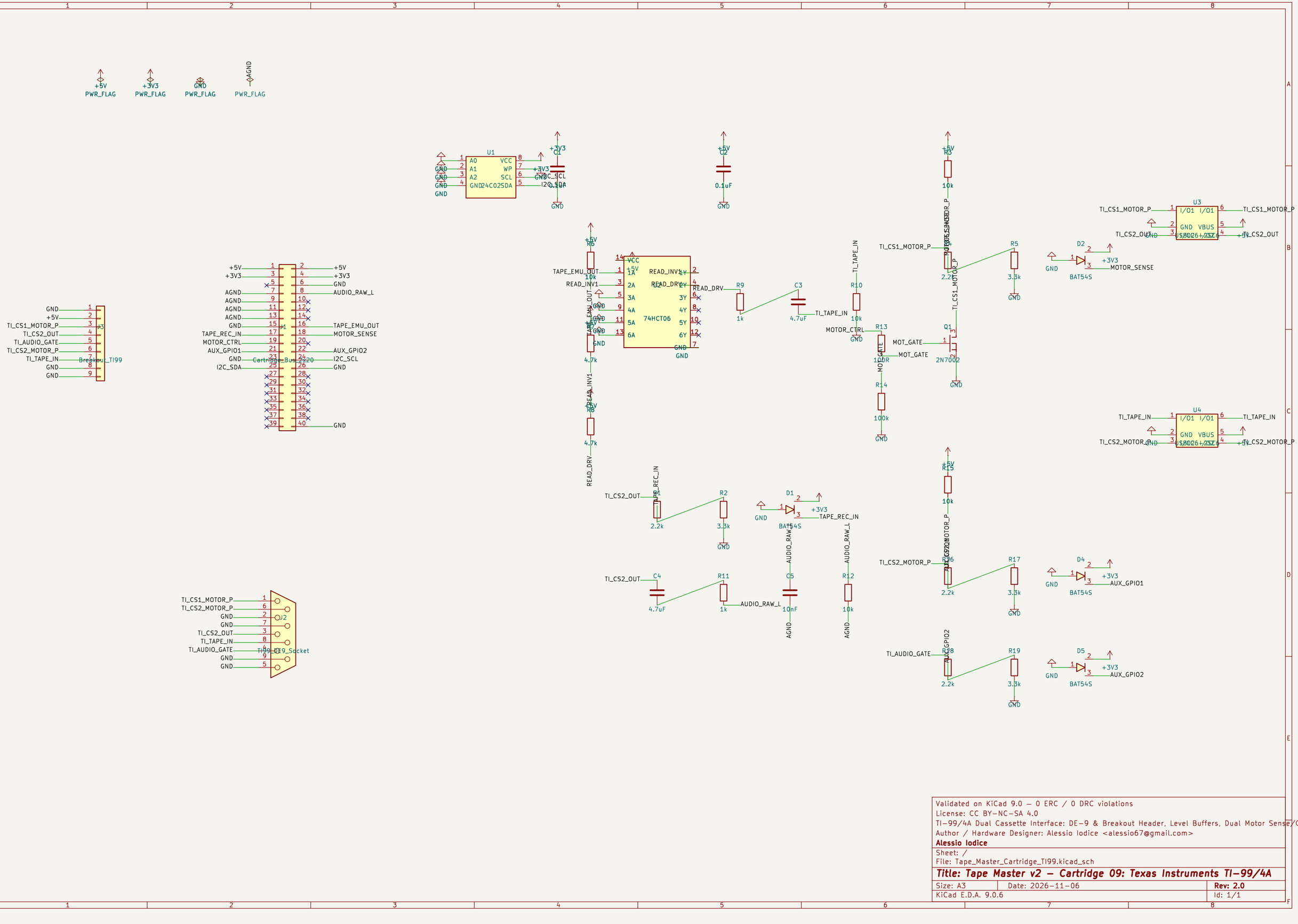

PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG
+5V
+3V3
GND
AGND
U1
A0
A1
A2
GND
VCC
WP
SCL
SDA
24C02
C1
0.1uF
C2
0.1uF
J3
Breakout_TI99
GND
+5V
TI_CS1_MOTOR_P
TI_CS2_OUT
TI_AUDIO_GATE
TI_CS2_MOTOR_P
TI_TAPE_IN
GND
GND
J1
Cartridge_Bus_2220
+5V
+3V3
GND
AUDIO_RAW_L
TAPE_EMU_OUT
MOTOR_SENSE
AUX_GPIO2
I2C_SCL
GND
AGND
TAPE_REC_IN
MOTOR_CTRL
AUX_GPIO1
I2C_SDA
J2
TI99_DE9_Socket
TI_CS1_MOTOR_P
TI_CS2_MOTOR_P
GND
TI_CS2_OUT
TI_TAPE_IN
TI_AUDIO_GATE
GND
GND
R6
10k
TAPE_EMU_OUT
R7
4.7k
R8
4.7k
READ_INV1
READ_DRV
74HCT06
R9
1k
C3
4.7uF
TI_TAPE_IN
R10
10k
MOTOR_CTRL
R13
100R
MOT_GATE
R14
100k
Q1
2N7002
R3
10k
TI_CS1_MOTOR_P
R4
2.2k
R5
3.3k
TI_CS1_MOTOR_P
D2
BAT54S
+3V3
MOTOR_SENSE
U3
TI_CS1_MOTOR_P
TI_CS2_OUT
I/O1
I/O1
GND
VBUS
USB026-252C
+5V
TI_CS1_MOTOR_P
TI_CS2_OUT
U4
TI_TAPE_IN
TI_CS2_MOTOR_P
TI_TAPE_IN
TI_CS2_MOTOR_P
TI_CS2_OUT
R1
2.2k
R2
3.3k
D1
BAT54S
TAPE_REC_IN
C4
4.7uF
R11
1k
AUDIO_RAW_L
C5
10nF
R12
10k
AUDIO_RAW_L
R15
10k
TI_CS2_MOTOR_P
R16
2.2k
R17
3.3k
D4
BAT54S
AUX_GPIO1
TI_AUDIO_GATE
R18
2.2k
R19
3.3k
D5
BAT54S
AUX_GPIO2
Validated on KiCad 9.0 — 0 ERC / 0 DRC violations
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TI-99/4A Dual Cassette Interface: DE-9 & Breakout Header, Level Buffers, Dual Motor Sense/Control,
Author / Hardware Designer: Alessio Iodice <alessio67@gmail.com>
Alessio Iodice
Sheet: /
File: Tape_Master_Cartridge_TI99.kicad_sch
Title: Tape Master v2 — Cartridge 09: Texas Instruments TI-99/4A
Size: A3
Date: 2026-11-06
Rev: 2.0
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Id: 1/1